



Recognize arcs

Find spline edges that a single circular arc represents within a tolerance -- report them, and optionally rebuild the wire with true arcs so Onshape shows their radius.

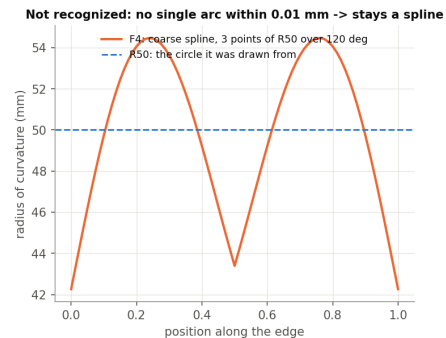


What it does

Imported and fitted curves carry arcs as splines: clicking them shows no radius, and drawings and CAM cannot use them as arcs. Recognize arcs asks of each spline 'is this really one arc?' and, where it is, puts a true sketch arc in its place.

Use it when

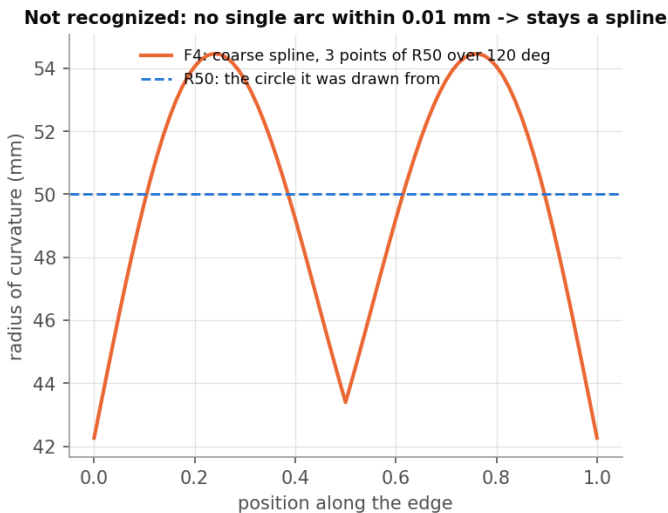
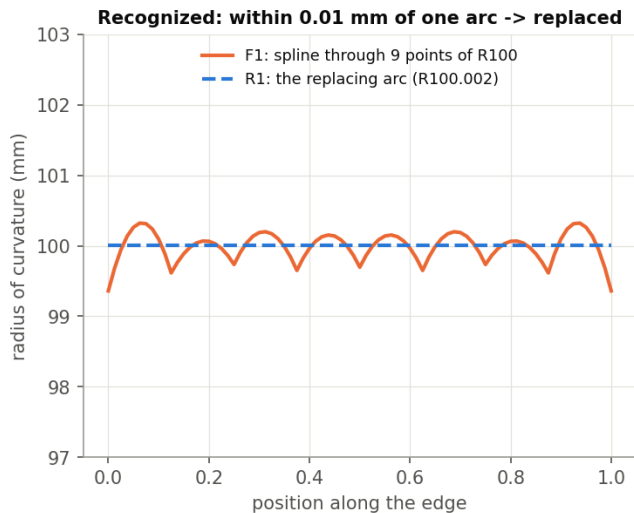
- A spline that should report a radius
- Check which edges of a wire are really arcs
- Before drawings or CAM that need arcs



R1: a spline through 9 points of R100 is replaced; R5's coarse spline is not.



The test



One arc, never a biarc

The circle through both ends and the best interior sample must be within tolerance at all 64 samples.

Ends agree

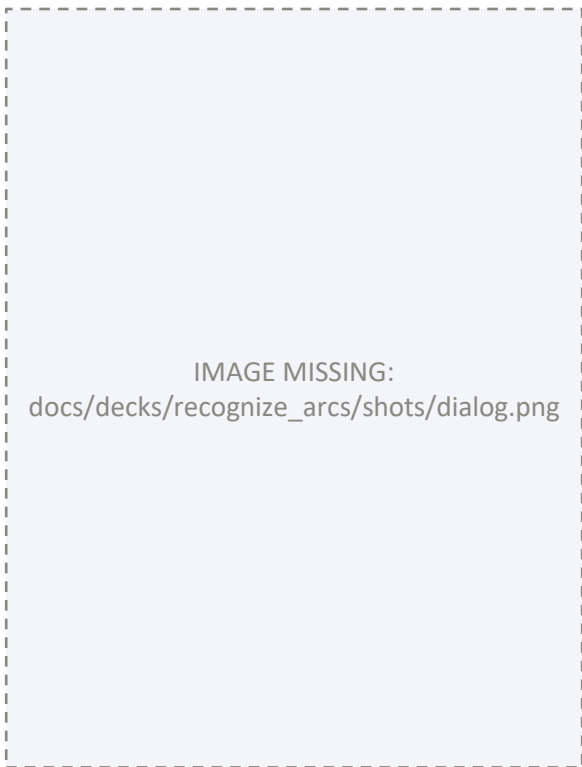
Both end directions within Max end tangent change of the circle's -- the kink a replacement can add.

Connected

The arc passes exactly through the edge's ends; lines and arcs pass through unchanged.



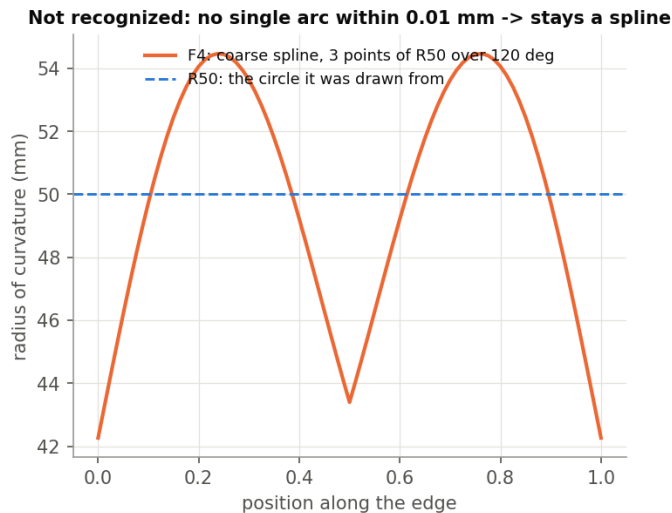
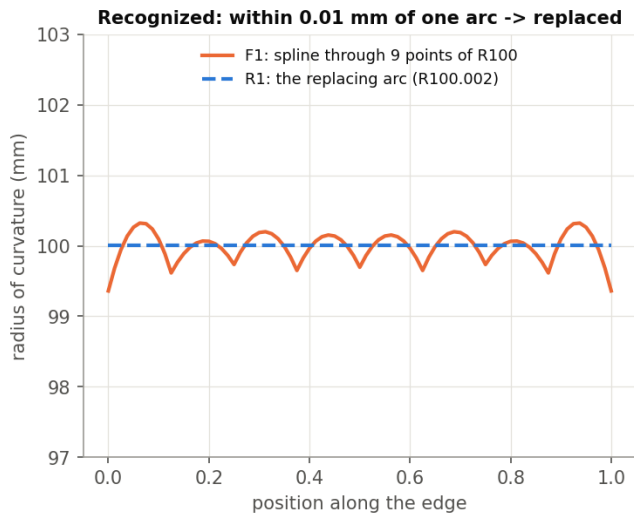
The dialog



#	Parameter	What it means
1	Edges or wires	The edges or wire bodies to examine (only splines are candidates).
2	Tolerance	Largest distance between the spline and its arc (0.01 mm).
3	Max end tangent change	Largest end-direction change allowed (0.05 deg).
4	Replace with arcs	On: build a copy with true arcs. Off: report and highlight only.
5	Name	Names the output wires.
6	Delete input	Delete input wire bodies once the copy is built.
7	Highlight matches	Show matching splines in green.



Example: a line-arc-chain (R2)



Line, spline on R200 (30 deg), line -> line, arc R200, line

One wire of 3 edges, joints within the 0.05 deg allowance. Also tested: R1 -> arc R100.002; R3 (S-curve) and R5 (coarse) stay splines; R4 report only -> no body.



Outputs and messages

Published for Extract variables

Key	What it holds
output	The rebuilt wire(s) (when Replace with arcs is on).
recognizedCount / candidateCount	Splines replaced / splines examined.
arcEdges	The new arc edges.

Messages

	Message / what to do
Info	N of M spline edges are arcs ... (deviation and tangent change per candidate) Near misses are visible.
Error	Select edges or wire bodies.



Tips and limits



Report first

Turn Replace off to see which edges qualify and by how much.



Kinks

A replacement can add up to Max end tangent change at a joint -- keep it small.



Where it lives

Document	Curve_tools (2143812a99089658c704f0bc)
Tab	recognize_arcs (uses curve_core)
Tests	Arc tangency tests: R1-R5 (9/9 incl. P12/P13, M1/M2)
Examples	Arc tangency tests studio
Explainer	docs/explainers/curve_tools/curve_tools_explained.md, section 2.7
Related	Clean wire, Map curve, Merge curve, Evaluate profiles, Fillet wire; Extract variables